

XRPL Validator — Daily Engineering Report • 2026-07-21

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- 11 commits (xrpl-validator 11)

Engineering changes

- **feat(node): differential_probe RPC disk cache — immutable historical lookups, DX_CACHE override** (a585ce8 , xrpl-validator)
- **feat(xrpl-ledger): real TicketCreate + OracleSet/OracleDelete transactors; tecDST_TAG_NEEDED — corpus 97.5%** (d28de50 , xrpl-validator) TicketCreate: batch ticket creation off tx seq (rippled CreateTicket.cpp shape), owner-dir spill via shared dir_insert, TicketCount/OwnerCount and Sequence landing past the batch. OracleSet: XLS-47 upsert with pair-level series merge and 1/2-unit reserve; new oracle keylet (0x0052 'R', mainnet-verified). Payment preclaim gains the lsfRequireDestTag check. Closes all 4 TicketCreate + 2 OracleSet SKIP-STUBs and the ...
- **feat(xrpl-ledger): real NFTokenPage model — mint/burn/accept/cancel/modify on structural page keys — corpus 98.2%** (454d6c6 , xrpl-validator) New ledger::nftpage: page key = owner(20) ++ low-96 bound, max page anchor, chain walk via PreviousPageMin, sorted insert, empty-page delete + relink. NFTokenMint composes the real NFTokenID (flags||feellissuer||taxon^scramble||seq off FirstNFTokenSequence+MintedNFTokens); AcceptOffer takes offer INDEXES (direct + brokered), moves the token between page sets, pays XRP with the id-embedded transfer fee, and clears offer ...
- **feat(xrpl-ledger): Number-exact rounding, XLS-80 domain books, multi-hop payment strands, real reserve — corpus 98.8%** (13197e6 , xrpl-validator) st_divide/st_multiply now round half-even folding the exact division remainder (the legacy +5/+7 truncation fudges double-rounded against the new nearest pass: 0EAE58BB fixed both ways, 42071037 proved the fudge had to go). DomainID (permissioned DEX) hashes into the book base for crossing AND placement — 105666804 replays 62/62. Payments execute multi-hop strands from the first path (chained book crossings, ...
- **feat(xrpl-ledger): full AMM lifecycle parity — Create/Deposit/Withdraw on real legs — corpus 99.1%** (a146364 , xrpl-validator) AMMCreate derives the pool account the rippled way (ripesha over sha512half(prefix0 || parentHash || ammKeylet), parent hash now threaded from the fixture header), creates the AMM AccountRoot + funds both legs via move_leg + mints creator LP tokens. Deposit/Withdraw move real legs between depositor and pool account and mint/burn the LP-token line (currency 0x03 ++ SHA512(minCur || maxCur)[0..19], mainnet-verified). ...
- **feat(xrpl-ledger): XRP autobridging for IOU ↔ IOU crossing; sender-reserve funding check — corpus 99.2%** (7111b1f , xrpl-validator) cross_bridged interleaves the direct book with the two-book XRP bridge (rippled FlowCross): peek both sources' head rates, consume one offer or bridge slice from the better one within

the taker threshold; the XRP intermediate rides through the taker and nets out of the mutation set. live_head only condemns unfunded offers whose backing state is actually loaded (phantom deletions poisoned attempt 1) and never mutates ...

- **fix(xrpl-ledger): me_muldiv 16-digit quotient scaling; AMM fee-adjusted spot; AMM turn in bridged crossing — corpus 99.3%** (7d51d97 , xrpl-validator) me_muldiv truncated to ZERO when the denominator mantissa dwarfed the numerator (XRP drops x IOU 1e-11 mantissas) — the bridge slice sizing died on it; the quotient now keeps ~16 significant digits. AMM spot quality embeds the trading fee (rate = in / (out·(1-f))) and the direct-pair pool competes inside cross_bridged, anchored to the best book rate. DX_BRIDGE/DX_AMM env diagnostics stay in. 105663160 replays 65/65; ...
- **feat(xrpl-ledger): rippled QualityFunction limit-trim for AMM fills — corpus 99.5%** (96dcd62 , xrpl-validator) Ported the exact single-path offer-crossing pipeline from rippled 3.2.0 (StrandFlow limitOut + BookStep post-check): when the AMM fill would beat the taker limit, the OUT request is trimmed via the pool QualityFunction (out = (1/q - b)/m under global Upward rounding, m/b at Near), re-swapped, and the ACHIEVED quality is re-checked with a 1e-7 relative tolerance granted only to trimmed requests. Directed roundings ...
- **feat(xrpl-ledger): issuer account-hops, endpoint dedup, in-flight hygiene — all corpus payments match — corpus 99.8%** (020c568 , xrpl-validator) Path account elements through the preceding hop's ISSUER are no-op re-anchors — accepted and skipped (true third-party rippling still unsupported). Endpoint currencies named as hops collapse via chain dedup. In-flight intermediates now leave NO trace: the sender's hop lines are restored byte-exact (Sandbox::forget for never-existed lines) and a mid-chain line-creation's directory droppings (both owners' roots + last ...
- **feat(xrpl-ledger): multi-path FIB AMM offers in bridged crossing — corpus stable 1107/1109** (f2d2bcb , xrpl-validator) IOU ↔ IOU crossings are always two strands on mainnet, so the direct-pair pool competes as rippled's generateFibSeqOffer: CLOB-like slices sized 0.025% of the pool's starting balances scaled by the Fibonacci sequence, judged by the slice's AVERAGE quality (spot+slippage+fee). Stops the pool from beating books it loses to by razor margins (A8830CA4 no longer touches the XAH/RLUSD pool). No corpus regressions.
- **feat(xrpl-ledger): per-leg AMM liquidity in bridged crossing — CORPUS 100% (1109/1109)** (c333d93 , xrpl-validator) Each bridge leg is a BookStep of its own pair, so its book head competes with that pair's pool via multi-path FIB slices — mainnet filled A8830CA4's XAH leg from the XAH/XRP pool while its RLUSD leg crossed the book, and BE7884AA's missing partial fill was the same arbitration. Leg sources resolve independently (book offer or pool slice), the XRP slice flows through whichever pair wins, and pool legs advance their ...

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Royalty Enforcement on NFT Escrow Transfers** (nft) Extend the XLS-51 NFTTokenEscrow object with the same royalty accounting NFTTokenAcceptOffer already performs: when an Escrow (ltESCROW) carries an sfNF

- **AMM-Triggered Stop and Take-Profit Offers** (dex) Add a tfTrigger flag to OfferCreate plus sfTriggerPrice (an STAmount ratio), sfTriggerDirection, and sfAMMID SFields on the Offer SLE, using the refer
- **Multi-Claim Pooled Escrow for Prize Pools** (gaming) Add a tfMultiClaim flag to EscrowCreate plus sfMaxClaimAmount and sfRemainingAmount SFields on the Escrow SLE. Under this flag, EscrowFinish no longer

Appendix — commits

Time	Repo	Commit	Subject
09:01	xrpl-validator	a585ce8	feat(node): differential_probe RPC disk cache — immutable historical lookups, DX_CACHE override
15:04	xrpl-validator	d28de50	feat(xrpl-ledger): real TicketCreate + OracleSet/OracleDelete transactors; tecDST_TAG_NEEDED — corpus 97.5%
15:19	xrpl-validator	454d6c6	feat(xrpl-ledger): real NFTokenPage model — mint/burn/accept/cancel/modify on structural page keys — corpus 98.2%
17:30	xrpl-validator	13197e6	feat(xrpl-ledger): Number-exact rounding, XLS-80 domain books, multi-hop payment strands, real reserve — corpus 98.8%
17:35	xrpl-validator	a146364	feat(xrpl-ledger): full AMM lifecycle parity — Create/Deposit/Withdraw on real legs — corpus 99.1%
17:57	xrpl-validator	7111b1f	feat(xrpl-ledger): XRP autobridging for IOU ↔ IOU crossing; sender-reserve funding check — corpus 99.2%
18:28	xrpl-validator	7d51d97	fix(xrpl-ledger): me_muldiv 16-digit quotient scaling; AMM fee-adjusted spot; AMM turn in bridged crossing — corpus 99.3%
18:56	xrpl-validator	96dcd62	feat(xrpl-ledger): rippled QualityFunction limit-trim for AMM fills — corpus 99.5%
19:11	xrpl-validator	020c568	feat(xrpl-ledger): issuer account-hops, endpoint dedup, in-flight hygiene — all corpus payments match — corpus 99.8%
19:30	xrpl-validator	f2d2bcb	feat(xrpl-ledger): multi-path FIB AMM offers in bridged crossing — corpus stable 1107/1109
19:39	xrpl-validator	c333d93	feat(xrpl-ledger): per-leg AMM liquidity in bridged crossing — CORPUS 100% (1109/1109)